

RIJU MARWAH

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Position Sought: ML Engineer, AgenticAI Engineer, ML Researcher

EDUCATION

Guru Gobindh Singh Indraprastha University
Bachelor of Technology, Computer Science Engineering
• GPA (Sem 6): 9.14, CGPA: 8.12

New Delhi, IN
Graduation Date: May 2026

EXPERIENCE

AI Institute, University of South Carolina, US
Research Intern

Remote/SC, USA
Mar 2025 - Present

- **Advisors:** Dr. Amit Sheth, Vishal Pallagani
- Studied AI fatigue and its correlation with hallucinations in large language models via novel frameworks.
- Programmed automated schema generation for structured input pipelines using LLMs.

EY (Ernst & Young)

AgenticAI Intern

Gurugram, IN
Jan 2025 - Mar 2025

- Built a low-code platform that reduced agent workflow development time by ~40%, using modular DAGs, LLM orchestration, and vector databases.
- Engineered task execution using Celery & Redis, handling 1,000+ concurrent tasks with retry and rate-limit logic.
- Integrated semantic agent routing, memory modules, and FAISS, improving agent accuracy by ~30%.

National Thermal Power Corporation

Summer Intern

Noida, IN
Jul 2024 - Aug 2024

- Developed ASP.NET Core MVC apps with Entity Framework, automating workflows and reducing manual processes by ~50%.
- Optimized MySQL and MongoDB CRUD operations using LINQ, enhancing query speed by ~25%.
- Secured apps with JWT auth, Google reCAPTCHA, and SMTP, reducing security risks by ~90% in testing.

PROJECTS & PAPERS

A Practical Toolkit for Identifying and Mitigating Cognitive Fatigue in LLMs

— *AAAI Demonstration 2026 (Submitted)*

- Introduced a novel fatigue-aware framework to detect and mitigate long-generation degradation in LLMs.
- Engineered per-token telemetry and a composite Fatigue Index to trigger real-time interventions
- Delivered a live HotpotQA demo across 4-bit Hugging Face models and decoders

Agentic AI Low-Code Workflow Platform (at Ernst&Young)

- Architected and built a low-code visual DAG composer for end-to-end AI workflows, enabling non-developers to drag-and-drop modular nodes (data prep, LLM prompts, vector retrieval)
- Developed a FastAPI backend to orchestrate Celery-powered tasks with Redis, ensuring fault-tolerant execution of complex, branching pipelines
- Incorporated OpenAI LLMs into node templates for dynamic prompt generation and response parsing, and enforced retrieval-augmented generation via Pinecone vector embeddings

Bhoomi, AI Powered Agricultural Productivity

- Built AI models for soil analysis and crop disease detection, delivering actionable insights to Indian farmers
- Crafted a Flutter app with transport booking, storage, market prices, multilingual chatbot for real-time support.
- Merged drone and mobile tech to boost yield and sustainability across diverse geographies.

MicroDetect-Net (MDN): Leveraging Deep Learning to Detect Microplastics in Clam Blood

— *Springer Nature, 8th International Conference on Innovative Computing and Communication (Link)*

- Achieved 92% accuracy in detecting microplastics from 276 fluorescence blood samples using CNN+UNET.
- Applied RGB thresholding and binary masking to isolate microplastics in fluorescence microscopy images.
- Deployed a real-time detection system using Flask, CNNs, and UNET for clinical research workflows.

TECHNICAL SKILLS

Languages: Python, C++

LLM & Agentic Frameworks: AutoGen, LangChain, CrewAI, OpenAI API, Function Calling, Prompt Engineering

ML & Deep Learning: scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Hugging Face Transformers

Data Processing & Analysis: NumPy, Pandas, Feature Engineering, Data Cleaning, Evaluation Metrics (Precision, Recall, AUC, RMSE)

Experimentation & Orchestration: Weights & Biases, Optuna, MLflow, Apache Airflow, Celery, n8n

Model Deployment & Infrastructure: FastAPI, Docker, REST APIs, TorchServe, TF Serving, Git

Vector Search & Retrieval: FAISS, Pinecone, Weaviate, OpenAI Embeddings

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Tools & Platforms: VS Code, PyCharm, LaTeX